I thank the Centers for Medicare & Medicaid Services for the opportunity to submit public comment on their Health Technology Ecosystem Request for Information. Below please find a letter w/attachment I recently emailed to HHS/CMS/CMMI requesting the opportunity to present Digitally Verifiable Adherence and Efficacy (DVAE) for every one of the approximately 400M doses prescribed to be self-managed every day by America's chronically ill population. Today the impact on health for even one of those doses is virtually unknown, resulting in up to $500B in avoidable costs and 275K deaths annually.

June 3, 2025

The Honorable Robert F. Kennedy Jr.
Secretary, U.S. Department of Health & Human Services

The Honorable Dr. Mehmet Oz
Administrator, Centers for Medicare & Medicaid Services

The Honorable Abraham Sutton
Deputy Administrator and Director, CMS Innovation Center

Subject: **Urgent: Digitally Verified Adherence & Efficacy (DVAE) - A Transformative Solution Aligning with CMMI's Strategic Objectives**

Dear Secretary Kennedy, Administrator Oz, and Director Sutton,

I am urgently requesting a brief video meeting to introduce a breakthrough that could close the most persistent and costly blind spot in American healthcare: the inability to verify whether self-managed medications are actually taken—and whether they work.

This breakthrough is **Digitally Verifiable Adherence and Efficacy (DVAE)**—a method that uses smartphone-connected packaging, wearable biosensors, and digital dispensing to confirm, in real time, both the adherence and the clinical impact of each medication dose taken by chronically ill patients. I have patented one approach, but I believe this field is ripe for explosive innovation—**if CMS/CMMI leads.**

Today, federal incentives unintentionally sustain a dangerous illusion. The Star Ratings' "Proportion of Days Covered" and the ACA's Medical Loss Ratio metric **reward the appearance** of adherence, not the reality. As a result, we pour billions into treating preventable disease progression caused by missed doses, undetected treatment failure, and medication mismanagement.

DVAE changes everything.

It transforms adherence from an unmeasured hope into a verifiable reality. It empowers patients with actionable feedback. It equips providers with real-world effectiveness data. And it gives CMS the power to unlock enormous cost savings through **tertiary prevention**—**right now.**

The enclosed Executive Summary details how DVAE directly accelerates each pillar of the CMS Innovation Center's **Strategy to Make America Healthy Again.** By verifying the most critical variable in chronic care—**whether patients are taking the right medications at the right time and whether those medications are working**—DVAE enables a future of radically improved outcomes, lower costs, and smarter value-based care.

But this future won't materialize unless CMS/CMMI acts.

I respectfully ask for 30 minutes to show how DVAE can be implemented with today's technology, and how modest policy levers—such as incorporating DVAE into Star Ratings or ACO quality incentives—can ignite market-wide adoption.

With leadership from CMS and CMMI, the U.S. can become the first country in the world where every self-managed dose is digitally confirmed and clinically meaningful. **This is the infrastructure chronic care has been waiting for.**

Thank you for your vision and leadership. I look forward to the opportunity to contribute to your mission.

Sincerely,
Stephen Mullen
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Enclosure: *Executive Summary – How DVAE Aligns with and Accelerates CMS/CMMI Strategic Objective*

How DVAE Aligns with and Accelerates CMS/CMMI Strategic Objectives

<u>To:</u> Centers for Medicare & Medicaid Services (CMS) / Center for Medicare and Medicaid Innovation (CMMI)
<u>From:</u> Stephen Mullen, Inventor of one DVAE method, info@intemed.net
Date: May 19, 2025

<u>Subject:</u> Digitally Verifiable Adherence and Efficacy (DVAE) — A Pillar-Enabling Innovation to Transform Chronic Care Management at Scale

The ability to **digitally verify the adherence and efficacy** (DVAE) of every self-managed medication dose taken daily by chronically ill patients—using widely available **smartphone technology**—offers CMS/CMMI the most direct, scalable, and cost-effective means to fulfill all three strategic pillars outlined in its innovation strategy.

DVAE as a Catalyst for CMMI's Strategic Objectives

1. Promote Evidence-Based Prevention – Tertiary Prevention at Scale

DVAE uniquely enables *tertiary prevention*—managing existing chronic conditions to prevent complications, hospitalizations, and decline—***by transforming the most consequential, yet least monitored, element of chronic care:*** **self-managed medication use**.

- **Precision Monitoring:** DVAE directly confirms whether patients are taking the right medication at the right time, and correlates this with biometric outcomes to assess real-world efficacy.
- **Immediate Cost Reduction:** Improved adherence alone reduces preventable ER visits, hospitalizations, and disease progression—generating rapid savings. Multiple studies estimate the current 50%+ non-adherence results in up to $500B in avoidable costs.
- **Evidence-Based Triage:** DVAE data allows early identification of ineffective treatments or deteriorating control, enabling proactive, evidence-based intervention.

2. Empower People to Achieve Their Health Goals

DVAE turns the smartphone into a real-time, personalized medication assistant—offering patients immediate feedback, secure tracking, and the confidence that their efforts are medically impactful.

- **Transparency and Accountability:** Patients and providers are empowered with verifiable data about whether the medication regimen is working.
- **Digital Literacy Inclusion:** DVAE integrates with existing mobile platforms and wearables, reducing friction and extending digital equity.
- **Integrated Goal-Setting:** Providers can tailor health goals and treatment plans based on real-time, individualized adherence and outcome data.

3. Drive Choice and Competition

DVAE democratizes chronic care data, allowing any provider—especially independent practices and community-based organizations—to participate in value-based care with high-fidelity insights.

- **Level Playing Field:** Smaller providers can demonstrate performance with DVAE metrics without relying on high-cost infrastructure.
- **Marketplace Differentiation:** Payers can offer better plans by incorporating DVAE-powered adherence data, enabling more accurate risk adjustment and targeted interventions.
- **Alternative Sites of Care:** DVAE supports at-home treatment with confidence, promoting site-neutral reimbursement and reducing facility dependence.

Why CMS/CMMI Must Lead the Industry to Implement DVAE

DVAE fills a critical blind spot in the current healthcare infrastructure: the ***invisible gap between medication prescription and outcome*****.** CMS/CMMI must act as the *motivating force* to integrate DVAE into the healthcare ecosystem for three urgent reasons:

1. **Systemic Market Failure:** No current regulatory or reimbursement model incentivizes verification of self-managed medication, ***despite its impact on >90% of chronic disease outcomes and costs.***
2. **Data Gap Undermines Value-Based Models:** Without verified adherence and efficacy data, value-based care, ACOs, MA plans, and capitation models ***cannot fully manage or price chronic care accurately.***

3. **Rapid ROI:** Unlike upstream prevention strategies that require years to show impact, ***DVAE enables immediate tertiary prevention***, which generates short-term cost savings and improved outcomes—aligning perfectly with CMS's mandate to protect taxpayer dollars.

Call to Action: Policy Levers CMS/CMMI Can Use to Motivate Adoption

To accelerate national implementation of DVAE, CMS/CMMI should:

- **Incentivize DVAE in Model Design:** Include DVAE capabilities as a condition for enhanced shared savings, quality scoring, or payment bonuses in ACOs, Medicare Advantage, and other value-based models.
- **Support Through Waivers:** Offer waivers to reimburse DVAE-enabling devices or apps as qualifying durable medical equipment (DME) under global risk models.
- **Tie to Star Ratings and Quality Metrics:** Require or reward DVAE-generated adherence and efficacy data in drug management metrics for MA and Part D plans.
- **Engage States and CBOs:** Integrate DVAE into Medicaid and CHIP pilots to support high-risk populations and promote digital inclusion.
- **Engage Stakeholders and the Private Sector**:
    - Partner with technology developers to refine DVAE platforms, ensuring accessibility for diverse populations (e.g., older adults, non-English speakers).
    - Collaborate with private payers to align DVAE incentives across multi-payer models, amplifying market demand and economies of scale.
- **Publicly Report on DVAE Impact:** Track and publicly report on the impact of DVAE implementation on adherence rates, clinical outcomes, and cost savings within CMS innovation models. This will build the evidence base and encourage wider adoption.

Conclusion

DVAE is not just an innovation—it is the ***missing infrastructure*** that allows CMS/CMMI to operationalize its bold strategic vision today. By verifying whether medication regimens are being followed and whether they work in real-world, self-managed environments, ***DVAE closes the last major gap in chronic disease management.***

With modest policy leadership, CMS/CMMI can enable a future where every medication dose taken by every chronically ill American is digitally confirmed and clinically meaningful—***fulfilling prevention, empowerment, and choice while safeguarding taxpayer funds like never before.***